FAT File System Cheatsheet v1.2

Master Boot Record

Offset	Desc	Length (byte)
000	Code area	446
1BE	1st Partition Table	16
1CE	2nd Partition Table	16
1DE	3rd Partition Table	16
1EE	4th Partition Table	16
1FE	MBR signature 0xAA55	2

Partition Table

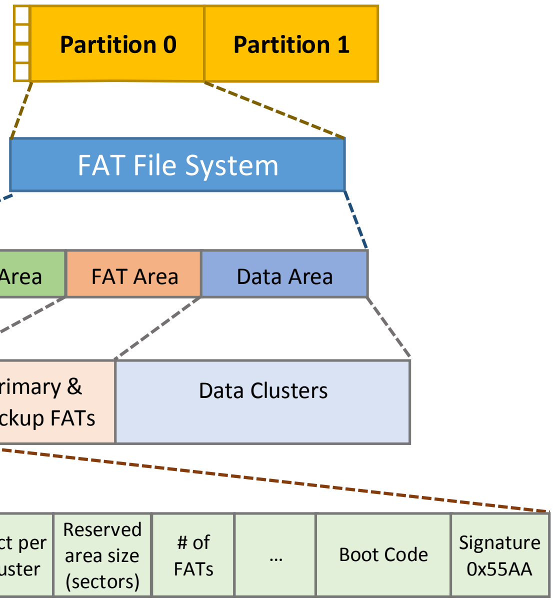
Byte	0	1-3	4	5-7	8-B	C-F
00 – 0F	Status	Start CHS	Type	End CHS	Start LBA	Size (sect)

CHS Encoding

Byte 1								Byte 2								Byte 3							
9				C				7				E				E				E			
1	0	0	1	1	1	0	0	0	1	1	1	1	1	1	0	1	1	1	0	1	1	1	0

1	0	0	1	1	1	0	0	1	1	1	1	1	0	0	1	1	1	1	0	1	1	1	0
9				C				3		E				1		E				E			
Head (8 bit)								Sector (6 bit)						Cylinder (10 bit)									

FAT File System Graphical Layout



CHS to LBA translation:

LBA = (((C * *cyl/head*) + H) * *sect/track*) + S - 1

FAT File System Layout

Reserved Area		FAT Area		Data Area		
Boot Sector	FSINFO [FAT 32]	FAT 0	FAT 1	Root Directory [**FAT12/16**: 32 sectors length]	Cluster #2 (FAT12/16)	*Rest of Data*
# of reserved sectors		# of FATs * sectors per FAT		(# of root entries * 32) / bytes per sector	# of clusters * sectors per cluster	

FAT12/16: Starting cluster # = 2

FAT32: Starting cluster #

→ found in boot sector (usually after FAT area)

FAT 12/16 Boot Sector

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	Boot code jump			OEM Name (ASCII)								Bytes per sect		Sect per cluster	Reserved area size (sect)	
10	# of FATs	Max # of root entries		16-bit value of # of sects		Media Type	FAT size (sect)		Sects per track		# of heads		# of sects before start of partition			
20	32-bit value of # of sects				BIOS INT13h #	Reserved	Ext. Bootsig	Volume serial number				Volume label (ASCII)				
30	Volume label (ASCII) -cont-						File system type label (ASCII)									
40	Boot Code															
…																
1F0															0xAA55	

FAT 32 Boot Sector

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	Boot code jump			OEM Name (ASCII)								Bytes per sect		Sect per cluster	Reserved area size (sect)	
10	# of FATs	Max # of root entries [0 for FAT32]		# of sects (16-bit) [0 for FAT32]		Media Type	FAT size (sect) [0 for FAT32]		Sects per track		# of heads		# of sects before start of partition			
20	# of sects (32-bit value)				32-bit FAT size (sect)				FAT struct index		Major/minor ver		Root dir cluster #			
30	FSINFO sect #		Backup boot sect #		Reserved											
40	BIOS INT13h #	Reserved	Ext. Bootsig	Volume serial number				Volume label (ASCII)								
50	Volume lbl (ASCII)-cont-		File system type label (ASCII)													
60 …	Boot Code															
1F0															0xAA55	

FAT Directory Entry

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	Filename [Byte 0 = 0xE5 or 0x00 for unallocated]								File extension			**File Attrib**	Reserved	Created Time (1/10s)	Created Time (hr,min,sec)	
10	Created Date		Accessed Date		1st cluster addr (high 2 bytes)		Modified Time (hr,min,sec)		Modified Date		1st cluster addr (low 2 bytes)		File Size (0 for directory)			

Long File Name (LFN) Directory Entry

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	Seq # [*]	Filename characters 1 to 5 (Unicode)										**File Attrib**	Reserved	Checksum	Filename char 6 (Unicode)	
10	Filename characters 7 to 11 (Unicode)										Reserved		Filename char 12 to 13 (Unicode)			

[*] OR-ed with 0x40 or contains 0xE5 if unallocated

File Attrib Flag Bits

Bit	5	4	3	2	1	0
Flag Set	Archive	Dir	Vol label	System File	Hidden File	Read only
			Long File Name (0x0F, 001111_2)			

Cluster to Sector Address translation:

Sect Addr = (Cluster# - 2) * (*sect/cluster*)
+ SectAddr of Cluster #2